

ANTONIO A. MALAGUTTI

CASE PORTFOLIO: PRODUCT OPERATIONS SPECIALIST

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🔍 1. ADVANCED PRODUCT DISCOVERY PRACTICES AND DUAL-TRACK AGILE

CASE #1

INSTITUTIONALIZATION OF AN ADVANCED CONTINUOUS DISCOVERY FRAMEWORK

Grupo Globo

CONTEXT (CHALLENGE)

Standardizing the continuous experimentation and discovery cycle to mitigate engineering and business risks prior to code development.

SITUATION

Squads spent excessive time refining ideas without prior user validation, leading to waste and software rework.

TASK

Design and scale a corporate Continuous Discovery framework connecting Senior Product Managers and designers.

ACTION

Implemented structured opportunity mapping routines (Opportunity Solution Tree) and validated hypotheses via vertical scope slicing.

RESULT

Significantly mitigated integration and business risks, accelerating the Time-to-Market of new validated features by 35%.

BUSINESS METRICS

Drastic reduction in rework caused by misaligned scope decisions

35% acceleration in Time-to-Market for validated features

TECHNICAL ARCHITECTURE

Integrated continuous research and feedback platforms

Central repositories for product hypotheses and experiments

GOVERNANCE

Strict corporate vertical slicing framework (Vertical Slicing)

Weekly hypothesis review rituals with Product Owners

CASE #2**STRUCTURING TECHNICAL AND BUSINESS DE-RISKING MATRICES**

Accenture (SolutionsIQ) / Accenture

CONTEXT (CHALLENGE)

Ensuring technical architecture viability and business value before the start of sprints in high-complexity ecosystems.

SITUATION

The highly iterative nature of technical AI refinements posed a severe risk of budget overruns and scope deviations.

TASK

Design rigid scope change governance mechanisms linked to early technical validation.

ACTION

Established technical validation gates tied to the Definition of Done (DoD) and instituted a baseline linked to data performance.

RESULT

The final budget variance was kept below 2%, shielding the profit margin and containing uncontrolled scope expansion.

BUSINESS METRICS

Final product budget variance kept below 2%

Fully protected and stabilized profit margins at 22%

TECHNICAL ARCHITECTURE

Data accuracy telemetry and infrastructure consumption tracking pipelines

Automated dashboards for workload threshold monitoring

GOVERNANCE

Definition of Done (DoD) criteria tightly coupled to minimum acceptable accuracy targets (85%)

Weekly budget governance committee with Burn Rate analysis

CASE #3**ORCHESTRATING VALUE VALIDATION BLUEPRINTS IN AI AND NLP**

PwC Brasil

CONTEXT (CHALLENGE)

Designing agile pipelines for hypothesis validation and legal compliance of complex data products.

SITUATION

Advanced Natural Language Processing (NLP) models faced indefinite deployment blocks due to regulatory risks.

TASK

Unify data scientists, compliance, and finance into an agile pipeline for the safe validation of AI products.

ACTION

Designed structural MLOps blueprints incorporating automated check gates and data anonymization routines.

RESULT

Achieved full approval from the risk and compliance audit, reducing the operational cycle time for document analysis by 70%.

BUSINESS METRICS

70% reduction in operational review cycle time

Zero security or non-compliance incidents during data auditing

TECHNICAL ARCHITECTURE

Continuous deployment and MLOps pipelines structured for advanced security

Automated encryption and personal data anonymization layers

GOVERNANCE

Multi-disciplinary sign-off gates involving AI, Risk, and Audit

Automated compliance gates integrated with LGPD/GDPR and ISO standards

CASE #4**DISCOVERY FLOW MAPPING VIA VALUE STREAM MAPPING (VSM)**

Natura / Accenture

CONTEXT (CHALLENGE)

Applying core Lean principles to identify wait states, hidden queues, and bureaucracy in the idea conception cycle.

SITUATION

The product demand pipeline suffered from chronic delays before reaching the engineering backlog due to manual handoffs.

TASK

Systematically map the value stream of discovery to eradicate process waste and accelerate flow.

ACTION

Conducted cross-functional Value Stream Mapping (VSM) workshops and automated redundant release approval gates.

RESULT

Achieved a 15% reduction in overall operational costs and a drastic optimization of the end-to-end product Cycle Time.

BUSINESS METRICS

15% reduction in operational process waste

Substantial compression of product delivery Cycle Time

TECHNICAL ARCHITECTURE

Analytical Value Stream visualization and modeling templates

Automated enterprise approval workflow systems

GOVERNANCE

Continuous improvement cadences based on Lean Digital mapping

Policies for waste elimination across the product demand pipeline

2. PRODUCT TELEMETRY, FLOW METRICS, AND ANALYTICAL DASHBOARDS

CASE #5

IMPLEMENTING FLOW TELEMETRY AND EMPIRICAL PREDICTABILITY VIA CFD

Grupo Globo / Accenture

CONTEXT (CHALLENGE)

Replacing subjective progress reports with mathematical and analytical data to govern the product lifecycle.

SITUATION

Squad status updates relied heavily on individual leader perceptions, failing to highlight queues and scope creep.

TASK

Implement an objective flow diagnostic system utilizing advanced engineering statistical tracking.

ACTION

Configured and monitored Cumulative Flow Diagrams (CFD), isolating expanding bands to neutralize operational bottlenecks.

RESULT

Eliminated subjectivity from scope negotiations, increasing delivery date estimation accuracy by 30%.

BUSINESS METRICS

30% boost in delivery date forecasting accuracy

Early diagnostic detection of expanding development queues

TECHNICAL ARCHITECTURE

Automated flow telemetry data collection via APIs

Real-time Cumulative Flow Diagram (CFD) analytical dashboards

GOVERNANCE

Weekly flow metric diagnostic reviews with engineering leaders

Data-backed scope negotiation policies based on empirical numbers

CASE #6**PYTHON STATISTICAL MODELING FOR DATA-DRIVEN BACKLOG****PRIORITIZATION**

Grupo Globo

CONTEXT (CHALLENGE)

Using specialized programming and data science to convert product usage logs into strategic priorities.

SITUATION

Backlog prioritization depended on political discussions and subjective feelings, resulting in a lack of focus on real value.

TASK

Build an objective automated pipeline to collect, parse, and cross-reference product usage technical metrics.

ACTION

Utilized Python and Pandas to extract raw data from APIs, consolidating Throughput and Defect Density indicators.

RESULT

Guaranteed assertive decisions backed by engineering data, driving a 15% increase in operational efficiency.

BUSINESS METRICS

15% increase in operational efficiency across prioritized areas

Reduction in product quality debt via objective numbers

TECHNICAL ARCHITECTURE

Automated scripts in Python for log consolidation

Metrics analytical parsing layer with Pandas via APIs

GOVERNANCE

Data-driven product decision-making policies

Bi-weekly operational velocity reviews backed by empirical telemetry

CASE #7**INSTITUTIONALIZATION OF WIP LIMITS AND LEAD TIME OPTIMIZATION**

insi (Meta IT)

CONTEXT (CHALLENGE)

Applying advanced Kanban mechanics to accelerate the throughput of product engineering pipelines.

SITUATION

Engineers suffered from multitasking overload, causing high context-switching costs and delayed releases.

TASK

Enforce strict operational flow management to compress the delivery Lead Time of new features.

ACTION

Enforced rigid Work-In-Progress (WIP) limits per column and trained teams on daily swarming tactics.

RESULT

Achieved an immediate 20% compression in end-to-end product Lead Time with complete throughput stabilization.

BUSINESS METRICS

20% compression in end-to-end product Lead Time

Drastic reduction in context-switching and operational waste

TECHNICAL ARCHITECTURE

Advanced Kanban analytical tracking systems

Automated alerts for stagnant pull request cycles

GOVERNANCE

Strict Kanban Management Professional (KMP) framework rules

Daily bottleneck-resolution and team swarming agreements

CASE #8**DESIGNING A UNIFIED HISTORICAL BASELINE FOR LONG-TERM ROI****MONITORING**

Claro / Accenture

CONTEXT (CHALLENGE)

Sustained evaluation of the financial return resulting from operating model shifts and product squad rituals.

SITUATION

Executive management questioned whether organizational transformation investments delivered genuine benefits or cosmetic noise.

TASK

Conduct a comprehensive, long-term study evaluating the actual ROI of process changes over six months.

ACTION

Systematically tracked cost-per-story delivery trends, throughput, and team lead times for six consecutive months.

RESULT

Empirically demonstrated a sustained 20% drop in internal operational costs, validating the operating model structure.

BUSINESS METRICS

Sustained 20% reduction in engineering operational costs

Empirical verification of the business value of process changes

TECHNICAL ARCHITECTURE

Unified historical delivery database layers

Long-term operational metric trend analysis tools

GOVERNANCE

Six-month continuous flow metric auditing

Post-transformation economic value calculation models

CASE #9

STANDARDIZATION OF REPOSITORY LIFECYCLES VIA SELF-SERVICE GOLDEN PATHS

Grupo Globo

CONTEXT (CHALLENGE)

Reducing cognitive load and accelerating team onboarding via standardized and automated infrastructure paths.

SITUATION

Dozens of squads faced weeks of delay trying to provision specialized environments and access credentials.

TASK

Design self-service 'Golden Paths' to accelerate the Time-to-Value of new analytical product tracks.

ACTION

Projected and delivered automated templates integrated with corporate password vaults for secure self-provisioning.

RESULT

Achieved full standardization of access governance for over 100 operational squads, reducing request cycles from days to seconds.

BUSINESS METRICS

Compression of analytical product Time-to-Value from days to seconds

Standardization of access governance for over 100 squads

TECHNICAL ARCHITECTURE

Data Platform as a Service (DPaaS) blueprints

Automated cloud infrastructure provisioning scripts

GOVERNANCE

Automated least privilege access control policies

Security verification integrated directly inside the delivery pipeline

CONTEXT (CHALLENGE)

Synchronizing complex portfolios, connecting C-Level strategy to the technical execution of engineering squads.

SITUATION

Chronic misalignment between corporate business priorities and the actual technical delivery capacity of software teams.

TASK

Implement the PI Planning ritual to tie complex OKRs to actual deployment capacity in structured batches.

ACTION

Led the facilitation of the scaled agile ritual, mapping critical inter-squad dependencies and designing the governance matrix.

RESULT

Guaranteed 100% predictability in regulated deliveries without budget overruns, paving stable development paths.

BUSINESS METRICS

100% alignment of technical deliveries to strategic OKRs

Elimination of budget deviations driven by scope creep

TECHNICAL ARCHITECTURE

Corporate software engineering capacity modeling

Inter-squad technical dependency mapping matrices

GOVERNANCE

Structured Program Increment (PI) Planning rituals

Continuous synchronization of delivery trains and System Demos

CASE #11**IMPLEMENTING LEAN PORTFOLIO MANAGEMENT (LPM) VIA WSJF****PRIORITIZATION**

Magazine Luiza / CI&T

CONTEXT (CHALLENGE)

Structuring a demand screening and governance framework to filter out low-value product initiatives.

SITUATION

An excessive volume of technology ideas entered the development pipeline without structured ROI analysis.

TASK

Establish a financial governance framework for demands to ensure engineering effort yields actual business return.

ACTION

Applied Lean Portfolio Management (LPM) practices using the WSJF (Weighted Shortest Job First) technique for backlogs.

RESULT

Achieved strict alignment of projects with strategic business objectives, optimizing Capex and eliminating scope waste.

BUSINESS METRICS

Maximization of technology portfolio Return on Investment (ROI)

Elimination of waste in product engineering feature creep

TECHNICAL ARCHITECTURE

Integrated Project Portfolio Management (PPM) systems

Corporate capacity planning tracking tools

GOVERNANCE

Lean Portfolio Management (LPM) framework practices

Demand prioritization and sequencing based on Cost of Delay (WSJF)

CASE #12**CENTRALIZING DATA PRODUCT GOVERNANCE VIA FEDERATED MODELS**

Grupo Globo

CONTEXT (CHALLENGE)

Strategic evolution of technical product management from decentralized models to managed platforms.

SITUATION

A decentralized Data Mesh architecture layout generated logical silos of duplicate data and high licensing costs.

TASK

Centralize operational data governance without breaking domain ownership autonomy across squads.

ACTION

Unified the architecture policies, managed cloud invoicing, and structured federated validation rituals.

RESULT

Massively optimized active computational infrastructure and simplified data consumption across the organization.

BUSINESS METRICS

Optimization and consolidation of corporate storage spending invoices

Drastic reduction in setup timelines for new analytics flows

TECHNICAL ARCHITECTURE

Integrated multi-cloud data architectures (AWS / GCP)

Decoupled data abstraction layers and software product infrastructure

GOVERNANCE

Centralized core corporate data templates and policies

Federated validation and compliance rituals for data domains



4. PRODUCT MATURITY EVOLUTION, CULTURE, AND CHANGE MANAGEMENT

CASE #13

ACCELERATING TEAM MATURITY VIA KOTTER AND ADKAR MODELS

insi (Meta IT)

CONTEXT (CHALLENGE)

Conducting major organizational product culture transitions for traditional retail clients.

SITUATION

Strong cultural resistance from traditional management tiers regarding the introduction of agile pipelines and new workflows.

TASK

Facilitate the structural cultural transition, mitigate governance friction, and accelerate squad maturity.

ACTION

Applied Kotter's 8-step model to build a coalition and the ADKAR model for individual awareness.

RESULT

Severely reduced political resistance, accelerating internal maturity and stabilizing sprint delivery pace.

BUSINESS METRICS

Exponential increase in adoption rates for new tools

Drastic reduction in voluntary turnover during transition

TECHNICAL ARCHITECTURE

Enterprise communication and digital training platforms

Internal engagement analytic mapping tools

GOVERNANCE

ADKAR change management framework

Application of Kotter's structured 8-step model

CASE #14**SQUAD EMPOWERMENT AND AUTONOMY THROUGH SERVANT LEADERSHIP**

Banco Original / CI&T

CONTEXT (CHALLENGE)

Driving team maturity by removing bureaucratic chains and coupling autonomy with metrics.

SITUATION

A culture of micromanagement and top-down approvals stifled technical creativity, slowing down release cycles.

TASK

Transition operations to a servant leadership model focused on metric-regulated autonomy.

ACTION

Replaced control directives with transparent product goals, removed technical blocks, and mentored squads.

RESULT

Significantly accelerated sprint velocity and predictability, yielding a sharp drop in engineering turnover.

BUSINESS METRICS

Substantial boost in sprint delivery velocity and predictability

Drastic reduction in voluntary engineering team turnover

TECHNICAL ARCHITECTURE

Decentralized autonomous squad infrastructure layout

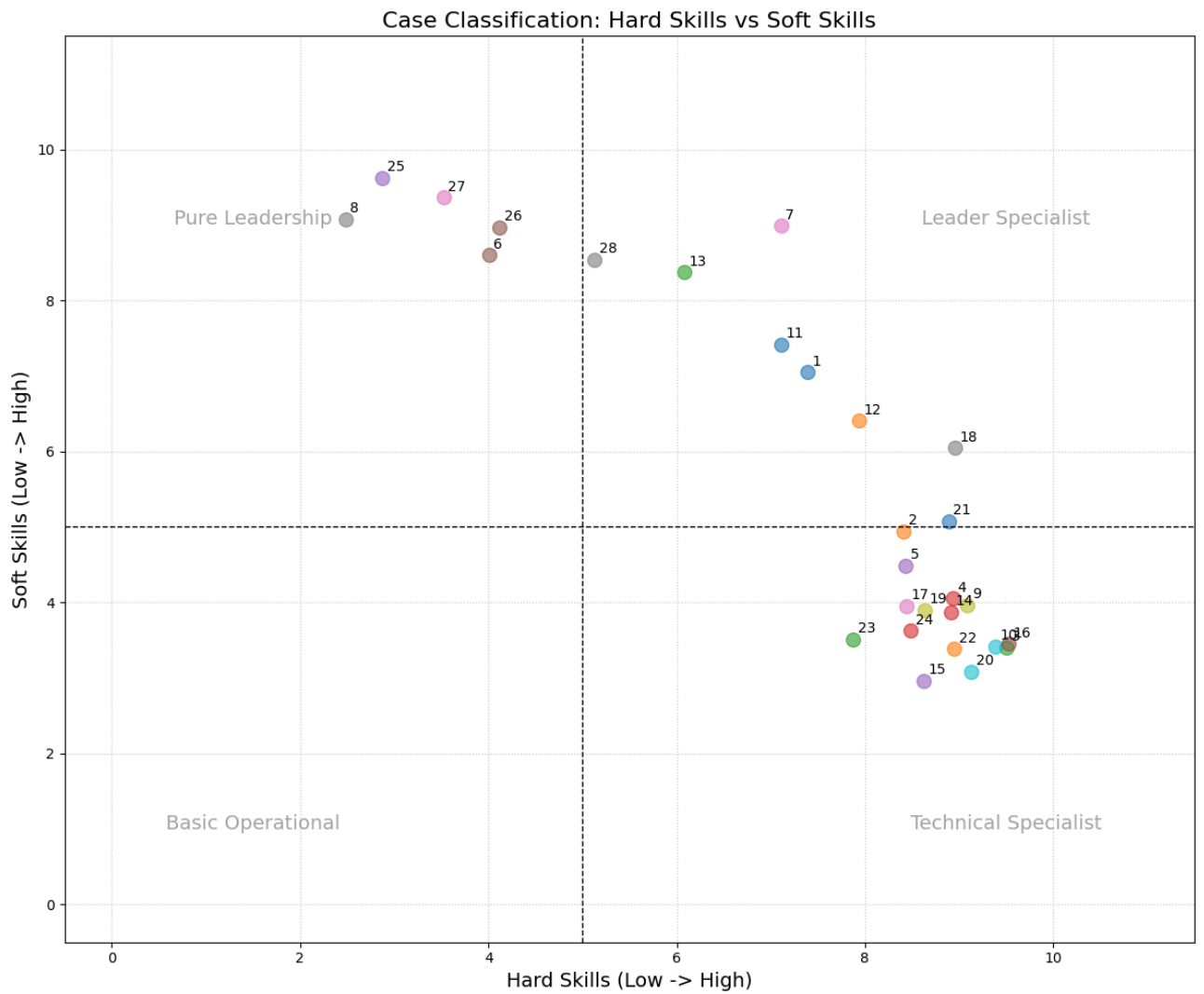
Automated team performance tracking boards

GOVERNANCE

Metric-backed squad autonomy and self-organization agreements

Cadences focused on the removal of operational blocks

CASE PORTFOLIO ANALYSIS: PROJECT MANAGEMENT



Visualization of the distribution of the 28 cases based on the balance between **Hard Skills** (Technical/Processes) and **Soft Skills** (Leadership/Resilience).

SITUATIONAL MEDIATION AND RESOLUTION OF CHRONIC FRICTION BETWEEN ENGINEERING AND QA

Grendene / Meta IT

CONTEXT (CHALLENGE)

Building high-trust environments by resolving long-standing friction between technical silos.

SITUATION

Intense internal disputes and a lack of synchronization between development and testing paralyzed production deploys.

TASK

Dissolve the operational deadlock, restore psychological safety, and reestablish release stability.

ACTION

Conducted conflict resolution workshops, created transparency spaces, and instituted shared quality targets.

RESULT

Completely repaired the working dynamic, generating release stabilization and a drop in post-deploy defect density.

BUSINESS METRICS

Sharp reduction in post-deployment defect density

Stabilization of sprint delivery predictability and velocity

TECHNICAL ARCHITECTURE

Integrated Dev-QA automated lifecycle metrics

Continuous integration and automated testing frameworks

GOVERNANCE

Conflict mediation and team psychological safety workshops

Shared quality KPIs balancing engineering output with testing rigor

STRATEGIC CAPABILITY DEVELOPMENT AND MENTORSHIP FOR TECHNICAL COMMUNICATION

FTD Educação / Accenture

CONTEXT (CHALLENGE)

Elevating the communication maturity of technical teams by translating structural debt into business risk.

SITUATION

Non-technical board members repeatedly rejected refactoring budgets because they did not understand engineering jargon.

TASK

Develop the ability within teams to translate technical debt into corporate financial risk language.

ACTION

Mapped decaying code blocks to clear business impacts, such as increased support costs and slower Time-to-Market.

RESULT

Secured full sponsorship and funding from the board for logical debt cleanup, stabilizing the platform.

BUSINESS METRICS

Secured 100% board funding for core engineering maintenance

Long-term drop in maintenance costs and platform stabilization

TECHNICAL ARCHITECTURE

Technical debt quantification metric models

Platform stability data visualization dashboards

GOVERNANCE

Jargon-free business risk communication frameworks

Strategic executive alignment and prioritization cadences

⚠️ 5. PRODUCT TURNAROUNDS, BUDGETS, AND CRISIS MANAGEMENT

CASE #17 STRUCTURAL PRODUCT TURNAROUND WITH CRITICAL SCOPE REDUCTION TO MVP

Mapfre Seguros / Accenture

CONTEXT (CHALLENGE)

Operational intervention in acute product crises to reestablish cost and timeline baselines.

SITUATION

A core insurance product faced a six-month delay and an entirely blown financial budget.

TASK

Stabilize the financial burn rate, restructure the chaotic backlog, and recover lost customer trust.

ACTION

Conducted an aggressive diagnostic, froze ongoing architectural changes, and cut the scope down to a lean, essential MVP.

RESULT

Successfully shipped the optimized MVP within a strict deadline, stabilizing finances and recovering client NPS.

BUSINESS METRICS

Financial stabilization of a distressed technology product budget

Complete recovery of customer confidence and Net Promoter Score (NPS)

TECHNICAL ARCHITECTURE

Temporary architectural modification freeze frameworks

Codebase quality and debt audit toolsets

GOVERNANCE

Total reset of the product operational governance model

Strict weekly checkpoint gates bound to scope reduction paths

CASE #18**EMERGENCY WAR ROOM SETUP FOR LIVE SYSTEM DOWNTIME MITIGATION**

Pernambucanas / Meta IT

CONTEXT (CHALLENGE)

Orchestrating emergency contingency plans when product system failures threaten digital revenue.

SITUATION

A primary transactional system crashed repeatedly due to chronic backend connection leaks, freezing sales.

TASK

Recover technical stability immediately and manage severe friction among executive stakeholders.

ACTION

Stood up a management-led emergency War Room, enforced rigid pooling limits, and isolated failing sessions.

RESULT

Achieved total platform stabilization within hours, reducing downtime by 90% and protecting live retail revenue.

BUSINESS METRICS

90% compression in critical system crisis downtime

Immediate protection and stabilization of live digital revenue streams

TECHNICAL ARCHITECTURE

Advanced Connection Pooling configuration in live environments

Live application activity monitoring infrastructures

GOVERNANCE

Daily management-led War Room structures

Post-incident root cause analysis and corrective roadmap creation

RECOVERING DELAYED PLATFORM MILESTONES VIA LEAN KAIZEN SOLUTIONS

Grupo Globo

CONTEXT (CHALLENGE)

Managing crises in corporate data products under intense pressure and accumulated quality debt.

SITUATION

A highly strategic data platform faced missing critical timelines due to hidden structural bottlenecks.

TASK

Take immediate control of the roadmap, isolate core impediments, and restore the delivery pace.

ACTION

Conducted intensive Kaizen workshops, re-mapped the immediate delivery flow, and focused engineering on high-value business items.

RESULT

Successfully rescued the release timeline, reestablishing stable delivery cadences and restoring C-Level trust.

BUSINESS METRICS

Successful recovery of critical delayed product milestones

Restoration of executive C-Level stakeholder confidence

TECHNICAL ARCHITECTURE

High-performance data platform infrastructure

Telemetria automatizada de rastreamento de débitos de qualidade

GOVERNANCE

Emergency Lean Kaizen continuous improvement cycles

Radical operational transparency and metric-based reporting

MEDIATION OF CROSS-FUNCTIONAL POLITICAL DEADLOCKS BETWEEN IT AND RISK AUDIT

PwC Brasil

CONTEXT (CHALLENGE)

Resolving structural conflicts and management standstills when control and technology areas collide.

SITUATION

A data integration bus project faced a complete block due to a priority clash with the corporate risk audit.

TASK

Resolve the political standstill, align stakeholder objectives, and secure executive backing for release.

ACTION

Facilitated targeted workshops to build shared goals and designed segmented communication tracks.

RESULT

Unlocked the political organizational deadlock, winning unanimous executive sponsorship for the final rollout.

BUSINESS METRICS

Resolution of executive deadlocks across distinct departments

Unanimous senior backing secured for high-impact milestones

TECHNICAL ARCHITECTURE

Stakeholder influence and interest tracking matrices

Segmented remote communication channels and alignment tools

GOVERNANCE

Shared-goal facilitation and alignment workshops

Hierarchically targeted tactical communication matrices

6. OPERATIONAL EFFICIENCY, PRODUCT FINOPS, AND BUDGETARY GOVERNANCE

CASE #21

AUTOMATED CAPEX/OPEX ACCOUNTING SEGREGATION BASED ON IT METADATA

Grupo Globo

CONTEXT (CHALLENGE)

Building product tracking mechanisms to shield corporate accounting against fiscal compliance risks.

SITUATION

The absence of clear tracing between hours invested in asset creation (Capex) and routine maintenance (Opex) generated compliance risks.

TASK

Build an objective, data-backed matrix to segregate engineering costs without manual, error-prone inputs.

ACTION

Tied development ticket metadata directly to accounting rules and monitored the weekly burn rate.

RESULT

Brought total audit clarity to the product budget for reviews, finishing the project 5% below the initial estimate.

BUSINESS METRICS

Final product delivery landed 5% under baseline budget

Complete mitigation of corporate financial compliance risks

TECHNICAL ARCHITECTURE

Integrated financial ledger and development tracking software

Automated cost dashboarding connectors

GOVERNANCE

Strict Capex/Opex product engineering segregation rules

Weekly ledger reviews matching engineering velocity to asset creation

CASE #22**PRODUCTIZING CLOUD FINOPS PRACTICES FOR MULTI-CLOUD INVOICING CONTROL**

Accenture / Accenture (SolutionsIQ)

CONTEXT (CHALLENGE)

Optimizing budgets by embedding infrastructure cost awareness directly into the product lifecycle.

SITUATION

Public cloud consumption within the ecosystem was scaling unchecked, eroding the business's operational profit margins.

TASK

Inject cost visibility into engineering and secure a target 25% reduction across cloud infrastructure budgets.

ACTION

Trained teams on cost-aware design patterns, implemented visibility tracks, and mandated right-sizing routines.

RESULT

Guaranteed a recurring 28% savings on cloud invoices without compromising system stability margins.

BUSINESS METRICS

28% recurring reduction in public cloud invoices

Substantial increase in overall cloud infrastructure ROI

TECHNICAL ARCHITECTURE

Integrated multi-cloud architecture layouts (AWS / GCP)

Automated cloud utilization and resource sizing profiles

GOVERNANCE

Institutionalized Cloud FinOps product policies

Quarterly cross-team cost and efficiency review cadences

CONTRACTS

Tetra Pak / CI&T

CONTEXT (CHALLENGE)

Preserving technical viability against client demands that exceed squad limits.

SITUATION

The client constantly demanded new features outside the contract scope, putting severe strain on the release schedule.

TASK

Firmly negotiate operational boundaries to defend team capacity without breaking the relationship.

ACTION

Used situational negotiation tactics backed by empirical team capacity data to contain scope creep.

RESULT

Maintained an optimal scope balance, shielding the squad from burnout and preserving NPS index baselines.

BUSINESS METRICS

Complete protection of core delivery timelines

Preservation of healthy client satisfaction indexes

TECHNICAL ARCHITECTURE

Detailed capacity metrics tracking software

Scope estimation validation databases

GOVERNANCE

Data-backed scope management guardrails

Situational negotiation and capacity boundary policies

EFFICIENCY MEASUREMENT ACROSS LARGE-SCALE GLOBAL IT CONTRACTS

CI&T

CONTEXT (CHALLENGE)

Using capacity planning data to manage the secure scaling of squad operational structures.

SITUATION

Commercial demands accelerating suddenly threatened to degrade delivery quality and breach contractual SLAs.

TASK

Scale up squad operations safely while keeping operational indicators and satisfaction high.

ACTION

Applied advanced capacity metrics to balance workloads and expanded the structure from 5 to 25 agile teams.

RESULT

Successfully expanded global contracts to 25 integrated squads, retaining high historical SLA and NPS baselines.

BUSINESS METRICS

Successful scaling from 5 to 25 high-performing teams

Sustained high customer satisfaction (NPS) and SLA metrics

TECHNICAL ARCHITECTURE

Global scale product team organizational infrastructure

Capacity planning metric tracking engines

GOVERNANCE

Strict SLA and contract compliance frameworks

Radical operational transparency and workload balance policies

7. TOOLING, PROCESS AUTOMATION, AND OPERATIONAL INTELLIGENCE

CASE #25

DEVELOPMENT OF INTERNAL AUTOMATION ENGINES VIA PYTHON SCRIPTS

MLGTT Consulting

CONTEXT (CHALLENGE)

Creating reusable custom classes and code modules to drive productivity gains in content workflows.

SITUATION

The analytical operational task of structuring image layouts and mosaics was manual, generating chronic bottlenecks.

TASK

Build a script utility to automate batch processing, compressing execution loops to minutes.

ACTION

Developed a specialized Python class utilizing Pillow and applying Lanczos scaling interpolation algorithms.

RESULT

Drastically cut the execution cycle from working days of layout design down to mere minutes, multiplying throughput.

BUSINESS METRICS

Drastic operational compression from working days to minutes

Exponential increase in internal asset processing capacity

TECHNICAL ARCHITECTURE

Python automated script engine layout

Pillow graphical processing library with Lanczos scaling

GOVERNANCE

Internal reusable script module policies (mlggt_utils)

Structured version control for automation assets

CASE #26**HANDS-FREE SANITY VALIDATION AUTOMATION FOR DATA REPOSITORIES**

MLGTT Consulting

CONTEXT (CHALLENGE)

Building custom preventive tools to ensure the reliability of massive corporate storage assets.

SITUATION

Daily corporate backups ran autonomously but lacked automated auditing routines against storage media corruption.

TASK

Develop a script utility to simulate restores and test logical sanity continuously.

ACTION

Wrote custom Python tools consuming specialized APIs to execute weekly automated restores.

RESULT

Created a perennially running, hands-free validation engine, shielding data assets against catastrophic failures.

BUSINESS METRICS

100% verified certainty of corporate data backup integrity

Elimination of manual database verification efforts

TECHNICAL ARCHITECTURE

Python automation scripts framework

Cloud Storage and database integrity validation libraries

GOVERNANCE

Strict data lifecycle continuity policies

Version-controlled automated utility script repositories (mlgtt_utils)

CONTEXT (CHALLENGE)

Sanitizing logical flows of transactional databases integrated with corporate analytical layers.

SITUATION

Inconsistencies during massive data extractions from industrial systems generated severe operational errors in BI.

TASK

Restructure the logical data pipeline by mapping foreign keys (Data Mapping) to ensure consistency.

ACTION

Led formal data cleansing tracks for orphan keys and designed optimized high-throughput flows.

RESULT

Stabilized analytical datasets at 100% reliability, generating a recurring R\$ 1.2M/year savings through technical efficiency.

BUSINESS METRICS

Audited recurring savings of R\$ 1.2 million per year

100% reliability achieved across core executive dashboards

TECHNICAL ARCHITECTURE

Enterprise ERP SAP integration backend

High-throughput ETL data pipeline architectures

GOVERNANCE

Dedicated data sanity and cleansing squad rituals

Strict transactional database integrity validation gates

CONTEXT (CHALLENGE)

Conducting experimental engineering projects to structure automated data monitoring capabilities.

SITUATION

The business required continuous, hands-free competitive intelligence gathering and market pricing data extraction.

TASK

Structure and code an experimental real-time web data scraping automation solution.

ACTION

Developed advanced web scrapers in Python leveraging specialized frameworks (Selenium and BeautifulSoup).

RESULT

Generated autonomous market intelligence under ethical compliance, unlocking new monetization channels and data products.

BUSINESS METRICS

Fully autonomous market intelligence gathering

Exponential increase in internal asset generation throughput

TECHNICAL ARCHITECTURE

Advanced web scraping mechanisms (Selenium / BeautifulSoup)

Structured data repositories for mass capture storage

GOVERNANCE

Agile cycle practices adapted for experimental Research and Development

Compliance guidelines and internal policies for ethical data scraping